Using Assessment Metadata to Quantify the Impact of Test Disengagement on Estimates of Educational Effectiveness

Jim Soland

NWEA

Oregon State University

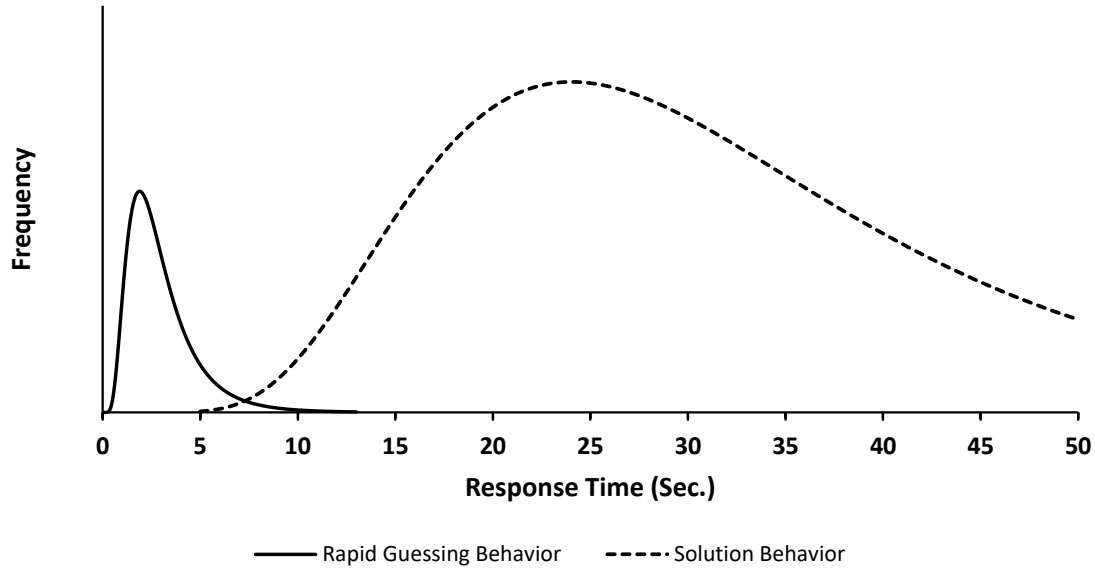
Read the sentence.

The grandfather clock has a metal pendulum.

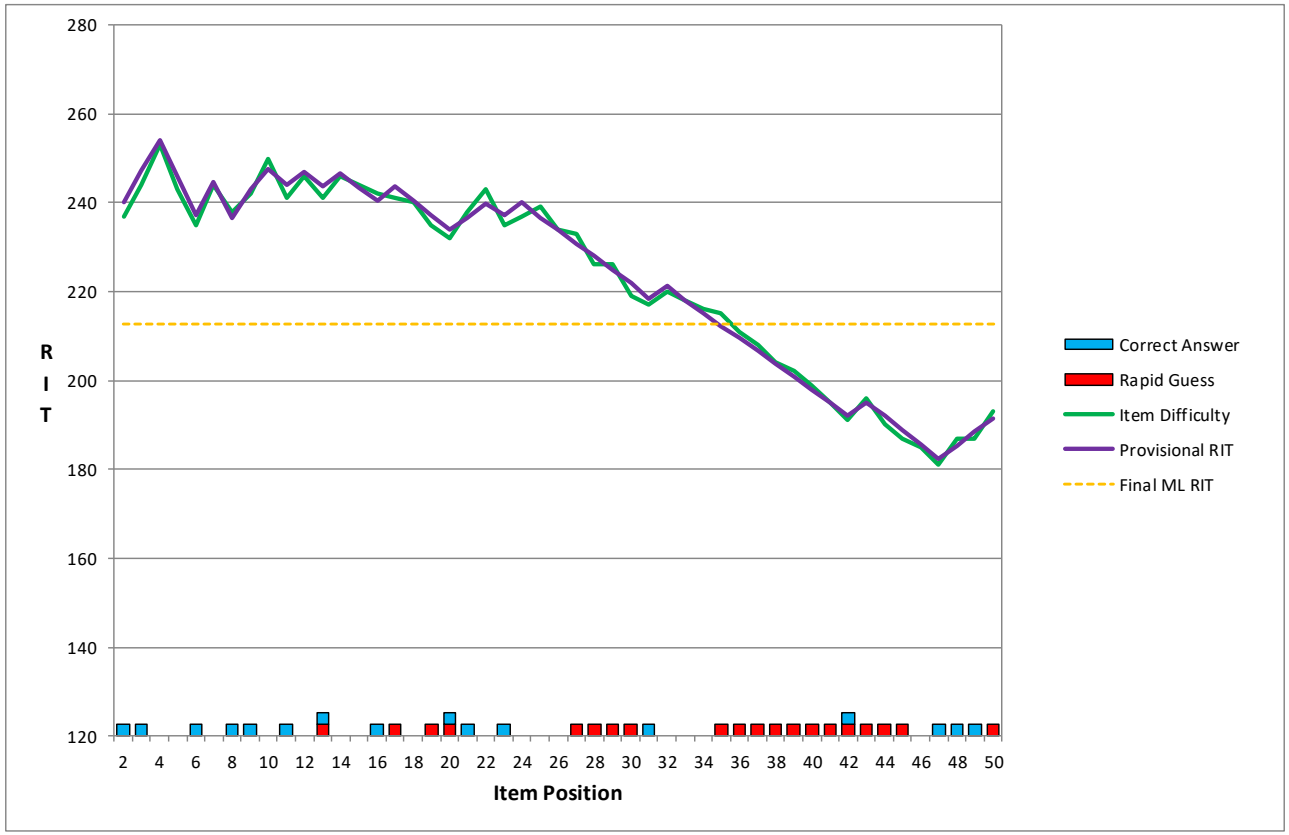
What is the meaning of the Latin root pend- in the word pendulum?

- A. to carry
- B. to hang
- C. to lead
- D. to pull

Next



Wise, S. L. (2017). Rapid-guessing behavior: Its identification, interpretation, and implications.


280
260
240
220
200
180
160
140
120
RIT
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
Item Position
Correct Answer
Rapid Guess
Item Difficulty
Provisional RIT
Final ML RIT

Two broad ways to address rapid guesses

- “Motivation filtering” (discussed by Rios et al., 2016)

- Item-level adjustments
  - E.g., the “effort-moderated score” (Wise & Kingsbury, 2017)

Research shows test disengagement can bias observed achievement scores

- Newer detection methods have been developed (e.g., Lee & Jia, 2014).
- Bias can be around .2 standard deviations under certain conditions (Rios et al., 2016).
- Magnitude of the bias dependent in part on whether the test is adaptive (Wise, 2015).
- These distortions can impact inferences using mean test scores (Jensen, Rice, & Soland, 2018; Soland, 2018).
- However, no real comprehensive look at how rapid guessing might impact common policy metrics.

Research Questions

Does rapid guessing affect estimates of

1. School effectiveness?
2. Achievement gaps, including changes in gaps?
3. Summer learning loss?

Table 1

Statistics on Analytical Sample

		Fall					Spring				
Year	Grade	N	Dis-engaged test events	RIT (un-adjusted)	RIT (Engaged test events)	RIT (Effort-moderated)	N	Dis-engaged test events	RIT (un-adjusted)	RIT (Engaged test events)	RIT (Effort-moderated)
2010-11	4	12,668	0.07	197.1 (15.6)	198.0 (15.3)	197.3 (15.4)	14,949	0.05	205.7 (14.6)	206.5 (14.1)	205.9 (14.5)
2011-12	5	16,945	0.07	205.0 (15.4)	206.0 (14.9)	205.2 (15.2)	18,883	0.05	211.7 (14.8)	212.6 (14.2)	211.9 (14.6)
2012-13	6	15,251	0.12	209.1 (15.6)	210.8 (14.8)	209.6 (15.3)	15,862	0.10	214.8 (14.9)	216.5 (13.8)	215.2 (14.5)
2013-14	7	14,262	0.20	213.9 (15.6)	216.2 (14.3)	214.5 (15.1)	14,729	0.10	218.8 (15.0)	220.5 (13.9)	219.2 (14.6)
2014-15	8	12,238	0.14	217.6 (15.7)	219.9 (14.5)	218.3 (15.2)	12,113	0.11	221.9 (15.5)	223.7 (14.3)	222.3 (15.0)

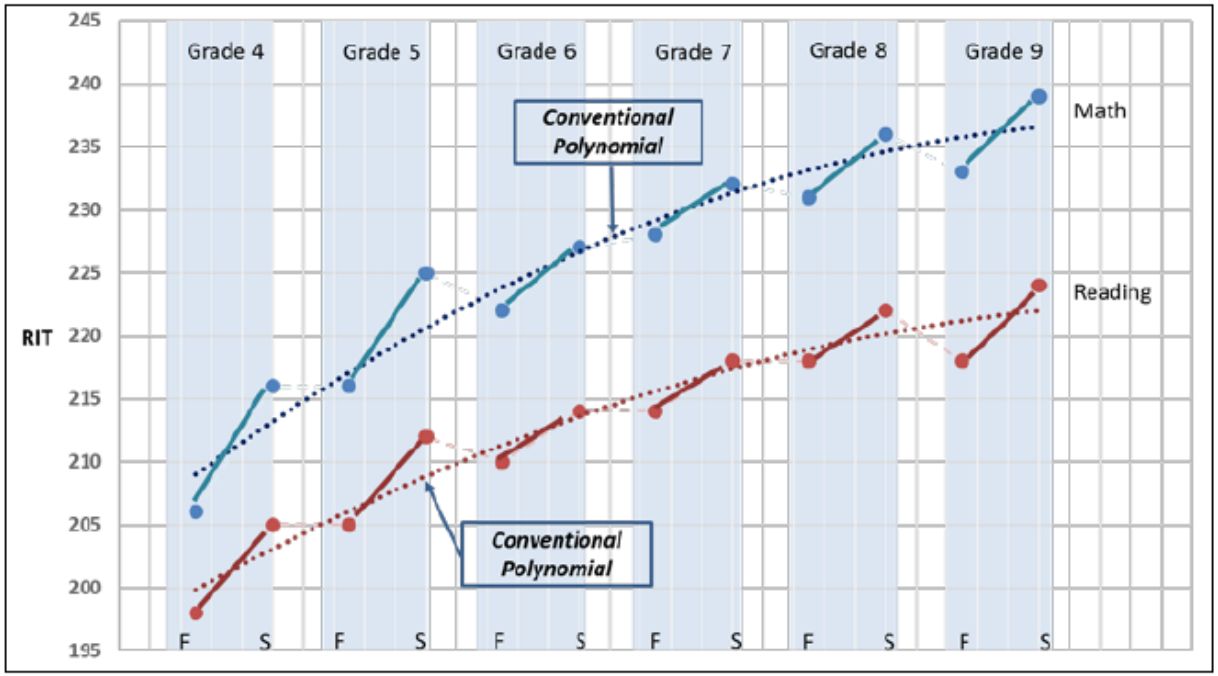
Note. Standards deviations are in parentheses. Disengaged test events occurred when a student rapidly guessed on 10% or more of the items.

Table 2

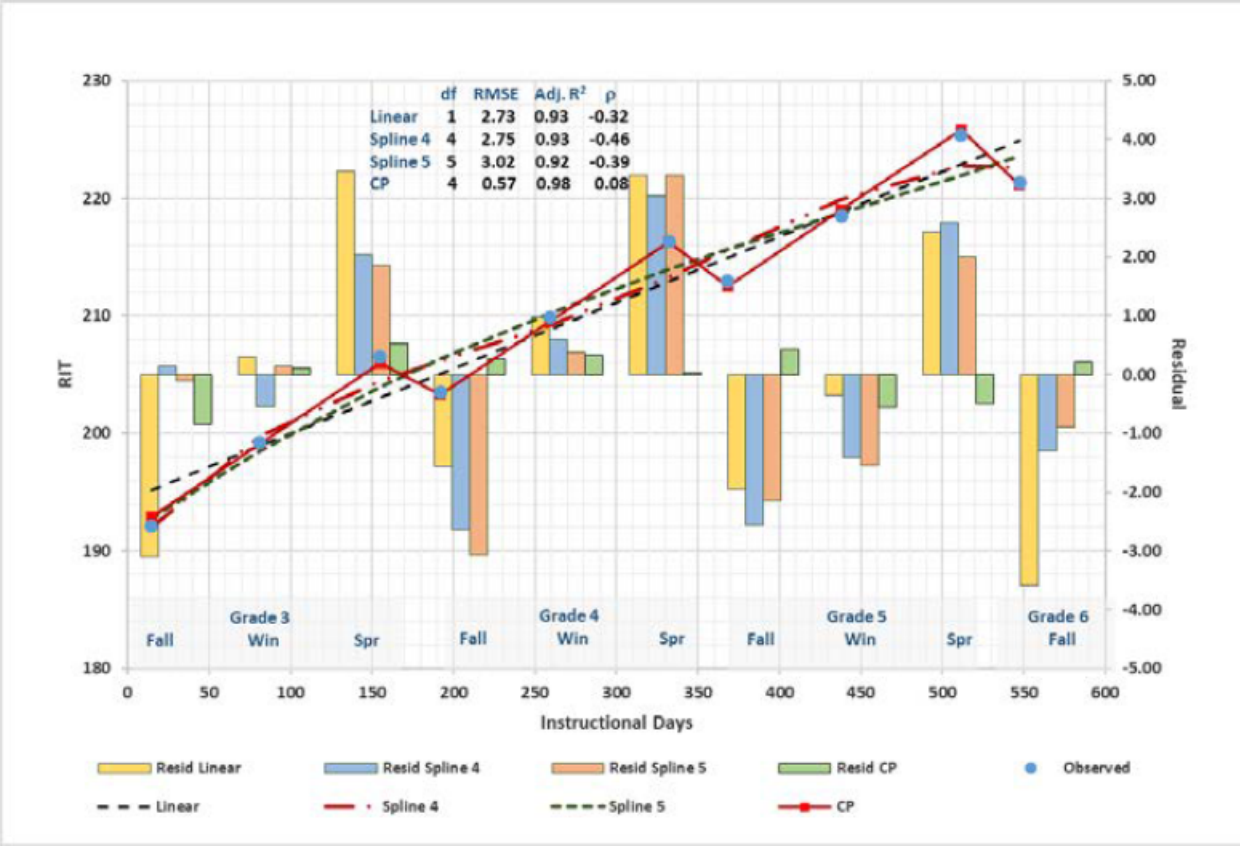
Proportion of Students with Disengaged Test Events (Rapid Guessed on 10% or More Items) by Gender and Race/ethnicity

Grade	Term	Year	White	Black	Asian	Hispanic
4	Fall	2010-11	0.06	0.09	0.03	0.05
5	Fall	2011-12	0.06	0.11	0.02	0.06
6	Fall	2012-13	0.11	0.16	0.03	0.12
7	Fall	2013-14	0.19	0.24	0.25	0.20
8	Fall	2014-15	0.13	0.17	0.03	0.15
4	Spring	2010-11	0.05	0.08	0.02	0.04
5	Spring	2011-12	0.04	0.08	0.02	0.04
6	Spring	2012-13	0.09	0.15	0.04	0.12
7	Spring	2013-14	0.10	0.14	0.01	0.10
8	Spring	2014-15	0.10	0.14	0.02	0.09

Compound Polynomial



Compound Polynomial



Compound Polynomial Specification

Equation 1. Level-1 Model (Repeated observations of MAP scores (t) within student (i) and school (j)):

$$\mathrm{y}_{tij} = \sum_{k=0}^{4} \pi_{kij} \mathrm{X}_{ktij} + e_{tij}$$

Compound Polynomial Specification

Level-2 Model (student (i) within school (j)):

$$\pi_{0ij} = \beta_{00j} + r_{0ij}$$
$$\pi_{1ij} = \beta_{10j} + r_{1ij}$$
$$\pi_{2ij} = \beta_{20j} + r_{2ij}$$
$$\pi_{3ij} = \beta_{30j} + r_{3ij}$$
$$\pi_{4ij} = \beta_{40j} + r_{4ij}$$

Compound Polynomial Specification

Level-3 Model (school (j)):

$$\beta_{00j} = \gamma_{000} + u_{00j}, \text{where } \gamma_{000} \text{ is the predicted Fall score at 4th grade}$$

$$\beta_{10j} = \gamma_{100} + u_{10j}, \text{where } \gamma_{100} \text{ is the linear growth rate of change of Fall scores}$$

$$\beta_{20j} = \gamma_{200} + u_{20j}, \text{where } \gamma_{200} \text{ is the quadratic growth rate of change of Fall scores}$$

$$\beta_{30j} = \gamma_{300} + u_{30j}, \text{where } \gamma_{300} \text{ is the predicted Fall} - \text{Spring growth in 4th grade}$$

$$\beta_{40j} = \gamma_{400} + u_{40j}, \text{where } \gamma_{400} \text{ is the linear growth rate of change for the Fall} - \text{Spring change}$$

Question 1: School contributions to student growth

Table 3

Estimates of School's Contributions to Student Growth Accounting for Rapid Guessing

	Fixed effects			Intraclass Correlation		
Parameter	No adjustment	Filtering	Effort-moderated score	No adjustment	Filtering	Effort-moderated score
Intercept	196.55 (0.43)	196.84 (0.42)	196.63 (0.43)	0.24	0.23	0.24
Fall-to-fall linear slope	6.25 (0.15)	6.67* (0.12)	6.54 (0.14)	0.37	0.27	0.32
Fall-to-fall quadratic slope	-0.35 (0.03)	-0.38 (0.02)	-0.37 (0.03)	0.35	0.31	0.32
Fall-spring change (4th grade)	7.58 (0.17)	7.60 (0.15)	7.63 (0.16)	0.60	0.45	0.54
Fall-spring change linear slope	-0.98 (0.06)	-1.02 (0.05)	-1.05 (0.06)	0.64	0.56	0.57

Note. Standards errors are in parentheses. * signifies difference between unadjusted and adjusted coefficients is significant at the .05 level.

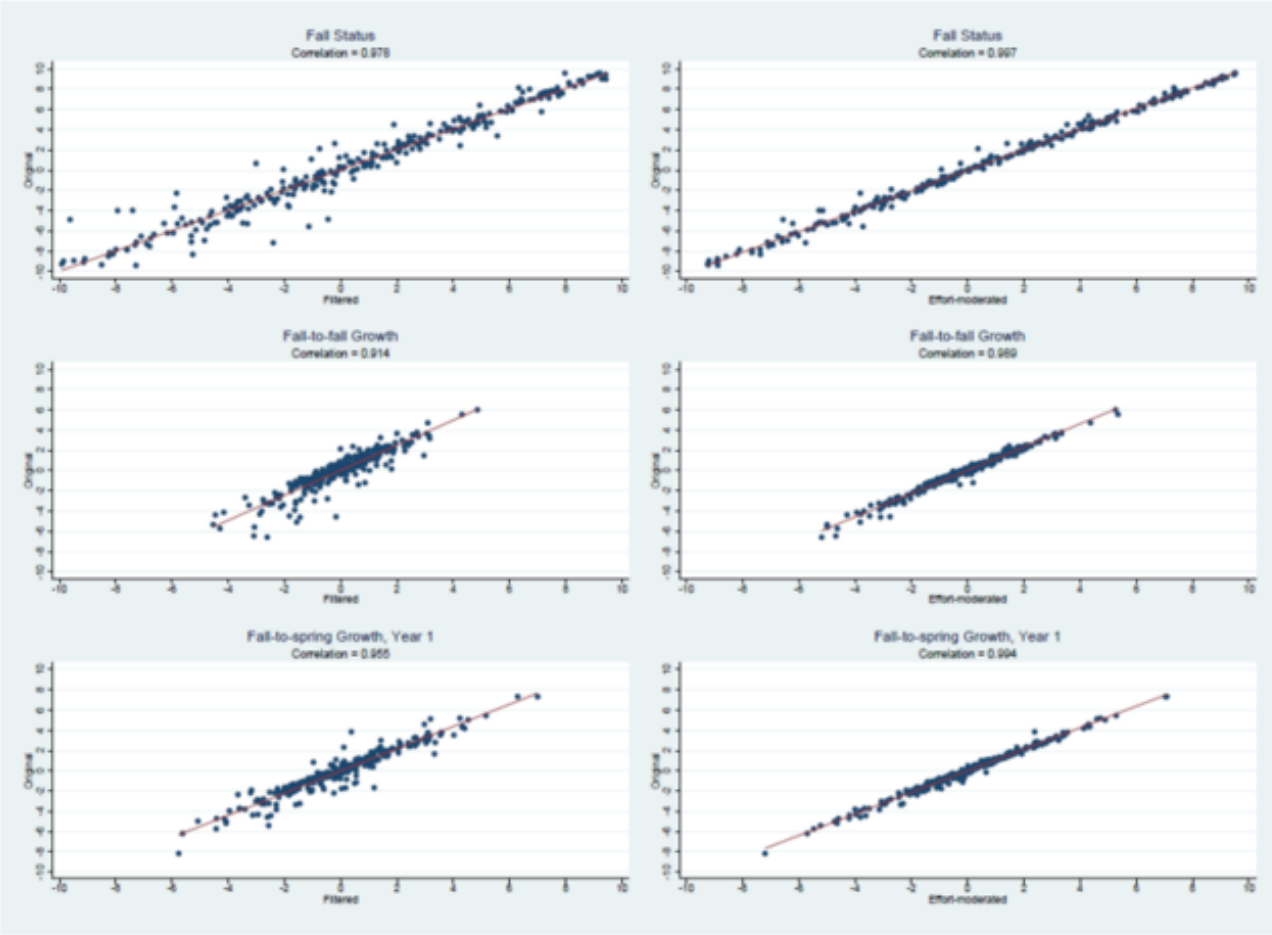


Figure 1. Comparisons of empirical Bayes estimates of school effectiveness.

Question 2: achievement gaps

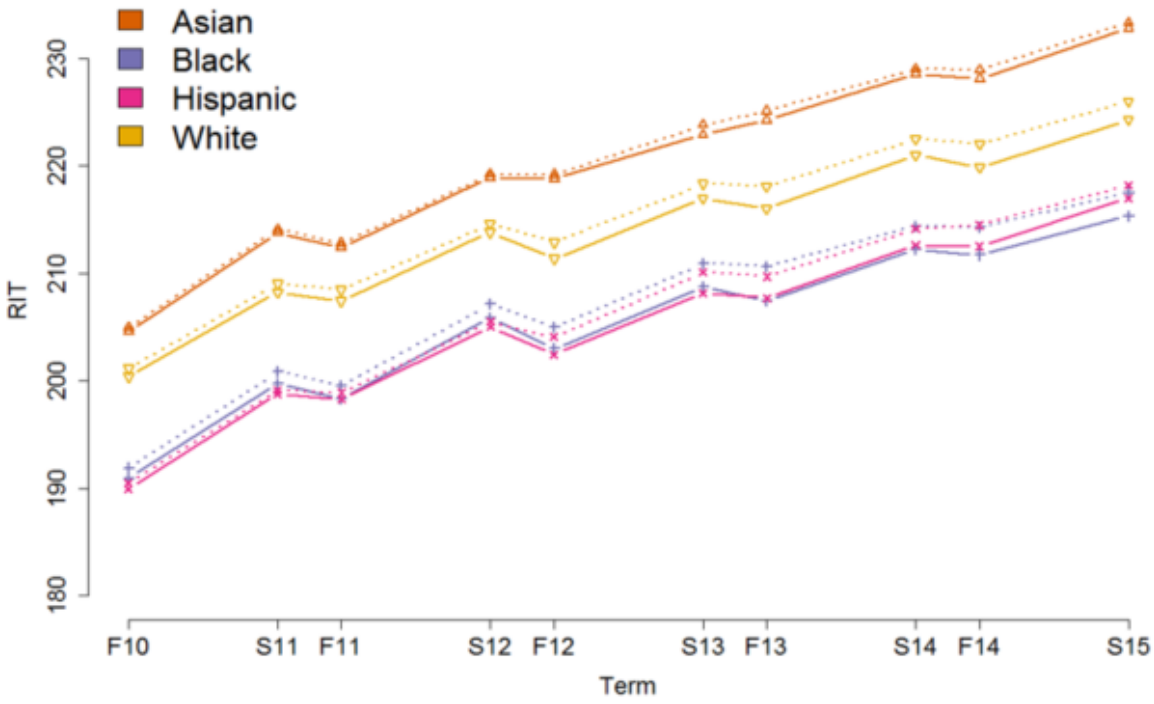


Figure 2. Comparisons of average student trajectories by race/ethnicity and whether filtering was used (where the solid line represents all students, and the dotted line represents only engaged students). Term represents the school term and year (F10 = Fall 2010). The dotted lines represent the trajectory of filtered score means for each race/ethnicity, and the solid lines represent the trajectory of unadjusted score means.

Table 4

Estimates of Racial/ethnic Gaps in Student Growth Accounting for Rapid Guessing

		Fixed effects		
Parameter	Group comparison	No adjustment	Filtering	Effort-moderated score
Intercept (4th grade fall)	White students	198.96 (0.39)	199.39 (0.37)	199.04 (0.39)
	Asian -White Difference	0.40 (0.89)	0.19 (0.91)	0.33 (0.90)
	Black -White Difference	-7.40 (0.41)	-7.70 (0.40)	-7.37 (0.40)
	Hispanic -White Difference	-5.47 (0.52)	-5.95 (0.50)	-5.56 (0.51)
Fall-to-fall linear slope	White students	6.24 (0.16)	6.47 (0.13)	6.47 (0.15)
	Asian -White Difference	0.95 (0.36)	1.09 (0.37)	0.95 (0.35)
	Black -White Difference	-0.06 (0.20)	0.43 (0.18)	0.13 (0.19)
	Hispanic -White Difference	-0.15 (0.26)	0.46 (0.22)	-0.03 (0.24)
Fall-to-fall quadratic slope	White students	-0.35 (0.03)	-0.35 (0.03)	-0.37 (0.03)
	Asian -White Difference	-0.02 (0.07)	-0.05 (0.07)	-0.03 (0.07)
	Black -White Difference	0.00 (0.04)	-0.08 (0.04)	-0.03 (0.04)
	Hispanic -White Difference	0.11 (0.05)	-0.02 (0.05)	0.09 (0.05)
Fall-spring change (4th grade)	White students	7.51 (0.18)	7.46 (0.16)	7.56 (0.17)
	Asian -White Difference	0.70 (0.30)	0.74 (0.30)	0.69 (0.30)
	Black -White Difference	-0.01 (0.19)	0.21 (0.19)	-0.01 (0.18)
	Hispanic -White Difference	0.28 (0.28)	0.39 (0.27)	0.32 (0.27)
Fall-spring change linear slope	White students	-0.95 (0.07)	-0.95 (0.06)	-1.01 (0.06)
	Asian -White Difference	-0.14 (0.13)	-0.20 (0.11)	-0.13 (0.12)
	Black -White Difference	-0.03 (0.09)	-0.14 (0.08)	-0.07 (0.08)
	Hispanic -White Difference	-0.08 (0.11)	-0.16 (0.10)	-0.12 (0.11)

Note. Standards errors are in parentheses. No significant differences were found between unadjusted and adjusted coefficients.

Question 3: summer learning loss

Table 5

Estimates of Summer Learning Loss Accounting for Rapid Guessing

	Fixed effects			ICC		
	No adjustment	Filtering	Effort-moderated score	No adjustment	Filtering	Effort-moderated score
4th Grade Spring Intercept	204.13 (0.42)	204.44 (0.41)	204.26 (0.41)	0.27	0.26	0.27
Fall-to-fall linear slope	5.27 (0.12)	5.65* (0.10)	5.49 (0.11)	0.34	0.25	0.29
Fall-to-fall quadratic slope	-0.35 (0.03)	-0.38 (0.02)	-0.37 (0.03)	0.35	0.32	0.32
4th-5th Summer change	-1.68 (0.12)	-1.31* (0.11)	-1.47 (0.12)	0.55	0.51	0.57
Summer change linear slope	0.28 (0.06)	0.26 (0.05)	0.30 (0.05)	0.42	0.48	0.41

Note. * signifies difference between unadjusted and adjusted coefficients is significant at the .05 level.

Some evidence of bias due to filtering

Table 6

Effort-moderated Reading Scores and Demographics by Proportion of Rapid Guesses, Spring 2015

Proportion of Rapid Guesses	Effort-moderated Score	N	Proportion of Students Who Are:				
			Female	White	Black	Asian	Hispanic
All	222.322	12,113	0.495	0.643	0.246	0.037	0.074
< .10 (unfiltered)	223.801	10,828	0.512	0.647	0.237	0.041	0.075
> .10 (filtered)	209.760	1,285	0.355	0.607	0.320	0.008	0.065

Note. The <.10 (unfiltered) scores were retained in the "filtering" analyses, while the filtered scores were excluded.

Discussion

- Generally, impact of rapid guessing on estimates of policy metrics is small
- Most of the impact is for filtering
- We have some evidence that filtering is introducing bias into the estimates by removing students with lower true scores
- However, one could certainly imagine scenarios where the robustness we see does not hold

Jim Soland

NWEA/Oregon State University

Twitter: @jsoland

E-mail: jim.soland@nwea.org

Soland, J. (In Press). Can the Amount of Time Students Spend on Test Items Help Illuminate Social-emotional Learning Needs? Initial Evidence from an International Achievement Test. *Educational Measurement: Issues and Practice.*

Soland, J. (2018). Are Achievement Gap Estimates Biased by Differential Student Test Effort? Putting an Important Policy Metric to the Test. *Teachers College Record*, 120(12).

Soland, J. (In Press). The Achievement Gap or the Engagement Gap? Measuring Black-White Achievement Gaps that Condition on Student Test-taking Engagement. *Applied Measurement in Education*.

Soland, J., Jensen, N., & Keys, T. (In Press). Is Low Test Motivation a Sign of Disengagement from School? Examining Indicators of Dropout Conditional on Rapid Guessing Scores. *Educational Assessment*.

Design Matrices

Appendix Table 1.
Coding of the Design Matrix for the Compound Polynomial model for Research Questions 1 and 2

Grade/Term	X_0 (Intercept)	X_1 (Fall linear slope)	X_2 (Fall quadratic slope)	X_3 (Fall-spring Int.)	X_4 (Fall-Spring linear slope)
4th Fall	1	0	0	0	0
4th Spring	1	0	0	1	0
5th Fall	1	1	1	0	0
5th Spring	1	1	1	1	1
6th Fall	1	2	4	0	0
6th Spring	1	2	4	1	2
7th Fall	1	3	9	0	0
7th Spring	1	3	9	1	3
8th Fall	1	4	16	0	0
8th Spring	1	4	16	1	4

Design Matrices

Appendix Table 2.

Coding of the Design Matrix for the Compound Polynomial model for Research Question 3

Grade/Term	X_0 (Intercept)	X_1 (Spring linear slope)	X_2 (Spring quadratic slope)	X_3 (Spring-Fall Int.)	X_4 (Spring-Fall linear slope)
4th Fall	1	-1	1	1	-1
4th Spring	1	0	0	0	0
5th Fall	1	0	0	1	0
5th Spring	1	1	1	0	0
6th Fall	1	1	1	1	1
6th Spring	1	2	4	0	0
7th Fall	1	2	4	1	2
7th Spring	1	3	9	0	0
8th Fall	1	3	9	1	3
8th Spring	1	4	16	0	0